

FOR OBJECTIVE CAUSAL INFERENCE, DESIGN TRUMPS ANALYSIS¹

BY DONALD B. RUBIN

Harvard University

For obtaining causal inferences that are objective, and therefore have the best chance of revealing scientific truths, carefully designed and executed randomized experiments are generally considered to be the gold standard. Observational studies, in contrast, are generally fraught with problems that compromise any claim for objectivity of the resulting causal inferences. The thesis here is that observational studies have to be carefully designed to approximate randomized experiments, in particular, without examining any final outcome data. Often a candidate data set will have to be rejected as inadequate because of lack of data on key covariates, or because of lack of overlap in the distributions of key covariates between treatment and control groups, often revealed by careful propensity score analyses. Sometimes the template for the approximating randomized experiment will have to be altered, and the use of principal stratification can be helpful in doing this. These issues are discussed and illustrated using the framework of potential outcomes to define causal effects, which greatly clarifies critical issues.

1. Randomized experiments versus observational studies.

1.1. *Historical dichotomy between randomized and nonrandomized studies for causal effects.* For many years, causal inference based on randomized experiments, as described, for example, in classic texts by [Fisher \(1935\)](#), [Kempthorne \(1952\)](#), [Cochran and Cox \(1950\)](#) and [Cox \(1958\)](#), was an entirely distinct endeavor than causal inference based on observational data sets, described, for example, in texts by [Blalock \(1964\)](#), [Kenny \(1979\)](#), [Campbell and Stanley \(1963\)](#), [Cook and Campbell \(1979\)](#), [Rothman \(1986\)](#),

Received May 2008; revised June 2008.

¹This work was supported in part by NSF Grant SES-05-50887 and NIH Grant R01 DA023879-01.

Key words and phrases. Average causal effect, causal effects, complier average causal effect, instrumental variables, noncompliance, observational studies, propensity scores, randomized experiments, Rubin Causal Model.

This is an electronic reprint of the original article published by the [Institute of Mathematical Statistics](#) in *The Annals of Applied Statistics*, 2008, Vol. 2, No. 3, 808–840. This reprint differs from the original in pagination and typographic detail.

Lilienfeld and Lilienfeld (1976), Maddala (1977) and Cochran (1983). This began to change in the 1970’s when the use of potential outcomes, commonly used in the context of randomized experiments to define causal effects since Neyman (1923), was used to define causal effects in both randomized experiments and observational studies [Rubin (1974)]. This allowed the definition of assignment mechanisms [Rubin (1975)], with randomized experiments as special cases, thereby allowing both types of studies for causal effects to be considered within a common framework sometimes called the Rubin Causal Model [RCM–Holland (1986)]. In particular, the same underlying principles can be used to design both types of studies, and the thesis of this article is that for objective causal inference, those principles must be used.

1.2. *The appeal of randomized experiments for estimating causal effects.*

For many years, most researchers have agreed that for drawing inferences about causal effects, classical randomized experiments, when feasible, are preferable to other methods [e.g., Cochran (1965)]. However, randomized experiments can be infeasible for a variety of ethical and other practical considerations, and the length of time we may have to wait for their answers can be too long to be helpful for impending decisions. Nevertheless, the possibility of conducting a randomized experiment should still be considered whenever a causal question arises, a point also made by Cochran (1965), which he attributed to earlier work by Dorn (1953).

Among the well-known reasons for this admiration for randomized experiments is the objectivity of the decisions for treatment assignment—the decision rules are explicit with a probability strictly between zero and one that each experimental unit will be exposed to either the treatment or control condition (for simplicity of exposition, this article will deal with the case of only two experimental conditions or exposures, called generically “treatment” and “control”). These unit-level probabilities, or propensity scores [Rosenbaum and Rubin (1983)], are known from the design of the experiment, and are all that are needed to obtain unbiased estimates of average treatment effects (i.e., the average effect of the treatment relative to control across all units). This unbiasedness property is suggestive of the powerful role that propensity scores play in causal effect estimation, even though unbiasedness is not an essential, or even always desirable, property of estimators.

Another reason why randomized experiments are so appealing, a reason, that is, of course not really distinct from their objectivity, is that they achieve, in expectation, “balance” on all pre-treatment-assignment variables (i.e., covariates), both measured and unmeasured. Balance here means that within well-defined subgroups of treatment and control units, the distributions of covariates differ only randomly between the treatment and control units.

A third feature of randomized experiments is that they are automatically designed without access to any outcome data of any kind; again, a feature not entirely distinct from the previous reasons. In this sense, randomized experiments are “prospective.” When implemented according to a proper protocol, there is no way to obtain an answer that systematically favors treatment over control, or vice versa.

The theme of this article is that many of the appealing features of randomized experiments can and should be duplicated when designing observational comparative studies, that is, nonrandomized studies whose purpose is to obtain, as closely as possible, the same answer that would have been obtained in a randomized experiment comparing the same analogous treatment and control conditions in the same population. In this process of design, the usual models relating observed final outcome data to observed covariates and treatment indicators play no part, just as they do not in the design of a randomized experiment. The only models that are used relate treatment indicators to observed covariates.

1.3. *Observational studies as approximations of randomized experiments.*

All statistical studies for causal effects are seeking the same type of answer, and real world randomized experiments and comparative observational studies do not form a dichotomy, but rather are on a continuum, from well-suited for drawing causal inferences to poorly suited. For example, a randomized experiment with medical patients in which 90% of them do not comply with their assignments and there are many unintended missing values due to patient dropout is quite possibly less likely to lead to correct inferences for causal inferences than a carefully conducted observational study with similar patients, with many covariates recorded that are relevant to well-understood reasons for the assignment of treatment versus control conditions, and with no unintended missing values.

The underlying theoretical perspective for the approach taken here was called the “Rubin Causal Model (RCM)” by [Holland \(1986\)](#) for a sequence of papers written in the 1970s [[Rubin \(1974, 1975, 1976a, 1977, 1978, 1979a, 1980\)](#)]. The RCM can be seen as having two essential parts, together called the “potential outcomes with assignment mechanism” perspective [[Rubin \(1990a\)](#), page 476], and a third optional part, which involves extensions to include Bayesian inference, only briefly mentioned here because our focus is on design, not analysis.

The first part of the RCM is conceptual, and it defines causal effects as comparisons of “potential outcomes” (defined in [Section 2](#)) under different treatment conditions on a common set of units. It is critical that this first part be carefully articulated if causal inferences are to provide meaningful guidance for practice. The second part concerns the explicit consideration of an “assignment mechanism.” The assignment mechanism describes the

process that led to some units being exposed to the treatment condition and other units being exposed to the control condition. The careful description and implementation of these two “design” steps is absolutely essential for drawing objective inferences for causal effects in practice, whether in randomized experiments or observational studies, yet the steps are often effectively ignored in observational studies relative to details of the methods of analysis for causal effects. One of the reasons for this misplaced emphasis may be that the importance of design in practice is often difficult to convey in the context of technical statistical articles, and, as is common in many academic fields, technical dexterity can be more valued than practical wisdom.

This article is an attempt to refocus workers in observational studies on the importance of design, where by “design” I mean all contemplating, collecting, organizing, and analyzing of data that takes place prior to seeing any outcome data. Thus, for example, design includes conceptualization of the study and analyses of covariate data used to create matched treated-control samples or to create subclasses, each with similar covariate distributions for the treated and control subsamples, as well as the specification of the primary analysis plan for the outcome data. However, any analysis that requires final outcome data to implement is not part of design. The same point has been emphasized in Rubin (2002, 2007) and the subsequent editorial by D’Agostino and D’Agostino (2007).

A brief review of the two essential parts of the RCM will be given in Section 2, which introduces terminology and notation; an encyclopedia entry review is given by Imbens and Rubin (2008a), a chapter length review is in Rubin (2008), and a full-length text from this perspective is Imbens and Rubin (2008b). Section 3 focuses on the assignment mechanism, the real or hypothetical rule used to assign treatments to the units, and on the importance of trying to reconstruct the hypothetical randomized experiment that led to the observed data, this reconstruction being conducted without examining any final outcome data in that observational data set.

Then Section 4 illustrates the design of an observational study using propensity scores and subclassification, first in the context of a classic single-covariate example from Cochran (1968) with one background covariate. Section 4 goes on to explain how propensity score methods allow the design of observational studies to be extended to cases with many covariates, first with an example comparing treatments for breast cancer to illustrate how this extension can be applied, and second, with a marketing example to illustrate the kind of balance on observed covariates that can be achieved in practice. Section 5 uses a Karolinska Institute example to illustrate a different point: that the same observational data set may be used to support two (or more) different templates for underlying randomized experiments, and one that may be far more plausible than the other. The concluding Section 6 briefly summarizes major points.